It's the Elicitation, Not the Eyes: Diagnosing VLM Failures in Slide Quality Inspection and Routing a Symbolic–Neural Critic

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Abstract

Vision–language models (VLMs) are now the default critic inside slide- and document-generation agents, yet they routinely pass slides whose text visibly overflows its box. Whether this reflects a failure of *perception* (the model cannot see the defect) or of *calibration* (it sees but will not commit) is rarely asked—and the answer decides whether the remedy is a larger model, a different prompt, or a different tool. We answer it with an attribution protocol built on a *lossless structured-geometry oracle* (element boxes, text, and style, rather than a model-written caption that re-injects perceptual error), and find that VLM "blindness" on rendered documents is two failures of opposite character. **Genuinely sub-perceptual** defects—fine alignment, font, color, margin—stay at chance across model scale (4B/8B/30B), five vision-encoder families, and two input resolutions, so they belong to a symbolic linter rather than a VLM. **Format-suppressed** defects only *look* sub-perceptual: the capability is present but the conventional pointwise+rubric prompt suppresses it, and simply *changing the elicitation*—a forced choice, or an atomic per-type query—recovers detection *with the model frozen*, a recovery that replicates across the capable models (four of the six VLMs tested, four families) and transfers to third-party human-annotated slides. This diagnosis prescribes the critic: a minimal *symbolic–neural hybrid*—linter for declared geometry plus one C3-prompted VLM for the rest—that covers 8/9 defect classes (balanced accuracy 0.896), matching the pre-registered routed hybrid's 8/9 coverage (0.885) where the best single engine covers 5/9. We sharpen the argument with a *falsifiable* linter-blind render class, G7 (declared box legal, rendered content overflows): a symbolic linter and two *narrow* neural rewards (a document and an aesthetic scorer, both at chance, 0.48 and 0.57) miss it, while a capable *general*-multimodal reward (0.79) catches it—so the blind spot is a property of narrow critics and G7 needs a VLM engine, whether re-elicited by prompt or carried by a reward head. The diagnosis itself reproduces on *real* CC-licensed slide layouts—a lossless tool-extracted oracle, no annotation—where fine alignment stays a capability floor in every channel while a margin bleed is structure-rescuable. As supporting evidence, an 8B examiner finetuned on injected defects surpasses a zero-shot 30B on *in-distribution* semantic inspection (the edge does not transfer out-of-distribution; Sec. 7), and a perturbation-fidelity audit shows that 45% of injected geometry defects never survive a standard renderer—a hazard any IR-labeled slide dataset inherits. We report the negatives plainly (deck-scope semantics, a sim2real gap, a small downstream effect); each sharpens rather than weakens the thesis.

1 Introduction

An automatic critic that flags layout and content defects is now a load-bearing component of every slide- and document-generation agent, and the default critic is a general-purpose VLM asked to grade a rendered slide. These critics fail in a revealing way: shown a slide whose title visibly spills out of its box, a strong VLM under a standard rubric prompt confidently answers "no defect." Whether this is a failure of *perception* (it cannot see the overflow) or of *calibration* (it sees but will not commit) is rarely asked—yet the answer decides whether the remedy is a larger model, a different prompt, or a different tool entirely. We treat that question as the design problem and answer it empirically.

The central observation. A VLM that "cannot see" a slide defect is hiding two failures of opposite character, and separating them is the whole game. Some defects are *genuinely sub-perceptual*: a three-pixel alignment offset or a

*Independent technical report. All experiments were run on a single 4×RTX 3080 (20 GB) workstation; weights, renders, and raw rollouts are not redistributed but all analysis scripts and reports are released. Correspondence: surh6@mail2.sysu.edu.cn.

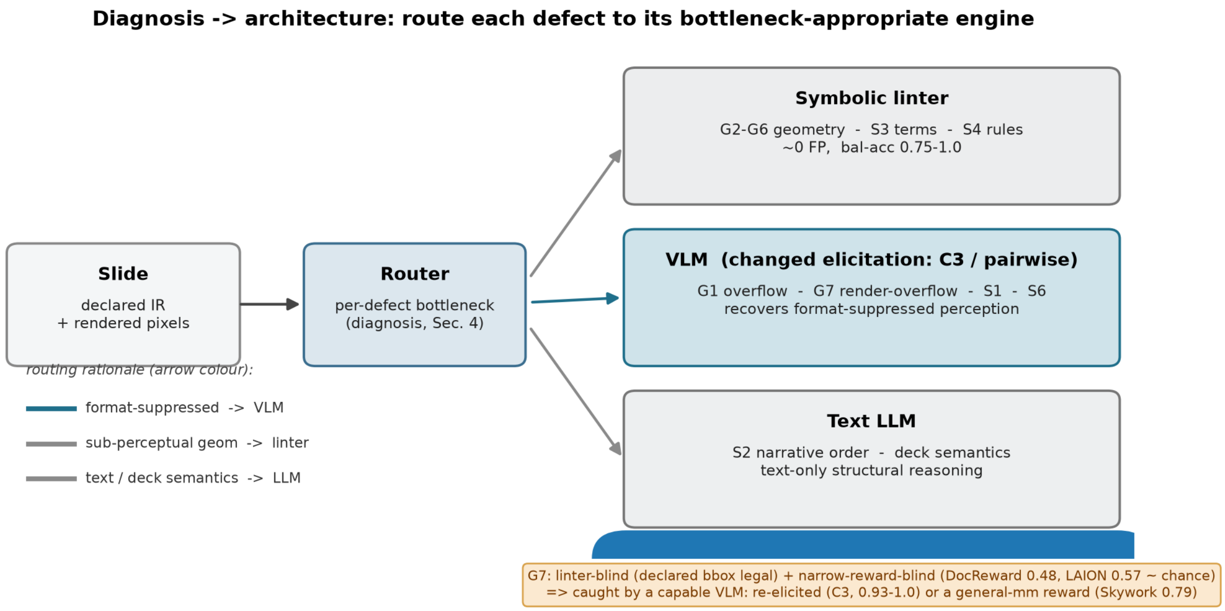

Figure 1: The paper in one picture. A perception/reasoning diagnosis (Sec. 4) shows slide-inspection failures are of two kinds—genuinely sub-perceptual (route to a symbolic linter) and merely format-suppressed (route to a VLM under a changed elicitation). Routing each defect to its bottleneck-appropriate engine yields a hybrid critic (Sec. 5). The render-overflow class G7 is the crux: a symbolic linter is blind to it by construction and *narrow* neural rewards (document, aesthetic) are blind to it empirically, while a capable general-VLM reward—and the VLM engine under atomic-binary elicitation—catches it, so G7 routes to a VLM engine.

one-point font mismatch sits below the model's effective acuity, and no amount of prompting, scale, or resolution recovers it. Others are *format-suppressed*: the model can perceive them, but the conventional elicitation—"here is a slide and a rubric; score every defect type in one pointwise pass"—buries the signal, and merely *changing how we ask* (a forced choice between two candidates, or an atomic yes/no per defect type with a required piece of evidence) recovers it with the model frozen. Conflating the two is why the field oscillates between "VLMs are bad at layout" and "VLMs can judge slides"—both hold, for different defects.

Diagnosis with a lossless oracle. We make the split measurable with an attribution protocol built on a *structured-geometry oracle*. Each slide carries an intermediate representation (IR) of element bounding boxes, text, and style; we present the same paired (defective, clean) slides under four input modalities—image-only, structured-oracle, VLM-caption, and image+oracle—crossed with detect/localize/repair tasks. Because the oracle is lossless machine-readable structure rather than a model-written caption, the perception channel it isolates is clean: a model that fails image-only but succeeds with the structured oracle is bottlenecked on perception, not reasoning. This is the methodological difference from concurrent perception/reasoning attribution that uses natural-language perception oracles, which re-inject the very perceptual error they try to factor out.

From diagnosis to architecture. The diagnosis prescribes the critic (Fig. 1): route each defect to the engine that owns its bottleneck. Sub-perceptual geometry goes to a symbolic linter that reads coordinates and rules at near-zero false-positive rate; text and structural semantics go to a language model; and the format-suppressed, perceivable classes go to a VLM, but only under the changed elicitation the diagnosis shows recovers them. To stress-test the design we construct an adversarial class, G7 render-containment overflow, that is invisible to the linter by construction (the declared box is legal; only the rendered pixels overflow) and—we find—invisible to *narrow* neural reward models (a document and an aesthetic scorer) as well, while a capable general-multimodal reward sees it, so it isolates exactly the slice of the problem that needs a VLM engine (prompted or reward-headed).

Contributions.

- **C1 – A perception/reasoning attribution protocol for generation quality-inspection**, built on a lossless structured-geometry oracle, that separates failures into *sub-perceptual* versus *format-suppressed* (Sec. 4) and reproduces on *real* CC-licensed slide layouts via a tool-extracted lossless oracle—no human or self-annotation (Sec. 7).
- **C2 – "Format suppression, not capability."** Relative and atomic-binary elicitation recover detection that the pointwise+rubric format suppresses; the contrast isolates format from capability, replicates across *the capable models—four of six VLMs tested, four families, three scales*, transfers to real human-annotated slides, and survives a localization check that rules out a lucky "yes" (Sec. 5).
- **C3 – A minimal symbolic–neural hybrid critic and a falsifiable linter-blind class.** The linter+VLM-C3 baseline covers 8/9 classes (0.896), matching the pre-registered routed hybrid (8/9, 0.885) and beating either single engine (linter 5/9, VLM-C3 4/9, VLM-C0 2/9); the render class G7 is caught only by a C3 VLM engine, while a symbolic linter (0.00) and two *narrow* neural rewards (a document reward 0.48 and an aesthetic scorer 0.57, both at chance) miss it—yet a capable *general*-multimodal reward catches it (0.79), so the blind spot is a property of *narrow* critics and G7 needs a VLM engine, prompted or reward-headed (Sec. 5).
- **C4 – A specialist examiner and a perturbation-fidelity audit (supporting).** An 8B examiner finetuned on injected defects surpasses a zero-shot 30B on *in-distribution* semantic inspection (it does not transfer out-of-distribution, Sec. 7) and learns to abstain rather than hallucinate geometry; separately, 45% of injected geometry defects never render under a standard template renderer—a hazard any IR-labeled slide dataset inherits (Secs. 6, 5).

This is a technical report, not a leaderboard entry. We have deliberately included the experiments that *cut against* the thesis—a near-zero downstream effect under a strong generator, a degenerate deck-scope head, and a synthetic-to-real transfer gap on bare pixels—because each one tells us something about where the perception/calibration boundary actually lies.

2 Related work

The VLM perception bottleneck. A growing line of work argues that VLM failures on visually grounded tasks are dominated by perception rather than reasoning—decomposing the bottleneck for chart understanding [1], or showing that the needed visual information is present in the representation but unused [2]; MMVP [3] is the canonical demonstration that VLMs stumble on minimally-different image pairs humans solve trivially—the sub-perceptual acuity floor our title ("not the eyes") deliberately echoes. We share the perception-first stance but ask it of a different task family—*quality inspection of generated artifacts*—where the perception channel can be isolated exactly with structured geometry rather than approximated by captions, and where the answer feeds a critic architecture rather than a benchmark.

Perception/reasoning attribution. Concurrent work separates perception from reasoning in abstract visual-reasoning benchmarks via natural-language perception oracles [4]. We differ in three ways: (i) our domain is inspection of generated documents, which admits a controllable defect taxonomy with *exact injected labels*; (ii) our perception oracle is a *lossless structured representation*, not a lossy caption, yielding a cleaner attribution boundary; and (iii) we close the loop from diagnosis to a deployed critic. Layout-error-detection work [5] pioneers distribution-grounded structural-error injection but for *document-layout-analysis predictions*, and explicitly notes that existing metrics cannot isolate whether failures stem from recognition or reasoning—precisely the gap our attribution protocol fills.

Relative vs. absolute judgement. Our finding that relative elicitation beats absolute scoring echoes work showing VLM judges can rank but cannot reliably score [6]; that result concerns a judge's *output mode*, orthogonal to our *source of failure*, and we adopt it to motivate the pairwise and atomic-binary heads of our critic. More broadly, pairwise/relative elicitation is known to recover discriminative signal that pointwise scoring suppresses in LLM/VLM judges; our contribution is not that effect but its *defect-class boundary*—it rescues G1/S6 yet not G3/S3—which is what turns the elicitation effect into a perception/format attribution signal.

Slide generation, evaluation, and reward. Slide and poster generation with design feedback is an active area; aesthetic-reward systems [7, 8] report that iterative VLM refinement helps gross layout damage but fails on fine-grained aesthetics and is prone to reward hacking, and slide-evaluation studies [9] call for calibrated critic-in-the-loop selection. These motivate, rather than compete with, a routed critic: they observe the VLM's perceptual ceiling and route around it with rules, whereas we first *diagnose* that ceiling and then show part of it is recoverable by elicitation. We use the human-annotated SlideAudit set [10] as our real-data probe and audit three published reward scorers—the document

Table 1: Defect taxonomy. The geometry group is detected symbolically; the semantic group is the examiner's domain; G7 is constructed to be linter-blind. "Channel" is the engine the diagnosis assigns (Sec. 4–5).

Group	Class	Description	Channel
Geometry	G1 text overflow	text exceeds its box	VLM (pairwise) + linter
	G2 element overlap	boxes intersect	linter
	G3 alignment offset	small mis-alignment	linter
	G4 font-size inconsistency	off-grid font size	linter
	G5 brand-color violation	off-palette color (ΔE)	linter
	G6 margin violation	element past safe margin	linter
Semantic	S1 title/body mismatch	body contradicts title	VLM
	S2 narrative-order break	deck sections out of order	LLM
	S3 terminology inconsistency	term drift across deck	term linter
	S4 density violation	slide over/under-packed	LLM
	S5 missing logic section	dropped argument step	LLM
	S6 image/text contradiction	figure contradicts caption	VLM
Render	G7 containment overflow	legal box, rendered overflow	VLM (atomic-binary)

reward DocReward [11], a general-multimodal value-head reward Skywork-VL-Reward [12], and a LAION aesthetic CLIP predictor [13]—as neural-critic baselines.

Critic / reward-model training. Training specialist critics from paired data is established for natural images and code; our examiner adapts the recipe to the rendered-document domain, where defect injection yields exact labels at zero annotation cost. We are careful *not* to claim injection itself as novel—it is a known recipe—only its use as the semantic engine of a routed critic and the perception/reasoning attribution it enables.

3 Problem setup and defect taxonomy

Representation and taxonomy. Every slide is an IR of typed elements with bounding boxes, text, and style; defects are injected programmatically into the IR and then rendered, giving exact labels and matched (defective, clean) pairs. The taxonomy (Table 1) has a *geometry/perception* group G1–G6, a *semantic/reasoning* group S1–S6, and one constructed render class G7; the taxonomy is aligned to the expert-derived SlideAudit taxonomy [10] via a bidirectional map (results are reported in its canonical classes), with G7 refining its "content overflow/cut-off" category. We add G7 because it is the sharpest test of the linter's blind spot: its declared box is legal (inside the safe margin, no box overlap) but the rendered content overflows its container, so the defect exists *only* in pixels.

Attribution modalities and metric. We present each paired sample under four modalities: **A** image-only; **B** the lossless structured oracle (with ground-truth markers stripped by a single de-leaking view, regression-tested); **B′** a VLM-written caption held fixed across models; and **C** image+oracle—crossed with detect/localize/repair. The attribution logic is: A-fail $\wedge$ B-succeed $\Rightarrow$ *perception* bottleneck; A-fail $\wedge$ B-fail $\Rightarrow$ *reasoning* bottleneck; A-succeed $\wedge$ repair-fail $\Rightarrow$ *execution* bottleneck. Throughout, the headline metric is *balanced accuracy on paired clean controls*, never recall alone—a consistency defect scored by recall reads a false "100% detection" when the model simply always flags.

Two linters. A geometry linter detects G1–G6 from coordinates and rules (brand color via CIELAB ΔE_{2000}, not RGB distance), at zero false positives on clean slides. A terminology linter handles S3 from an occurrence table plus near-duplicate clustering, image-free. Both are the "detectors of record" the diagnosis routes to.

4 Diagnosis: two kinds of blindness

Fine geometry is genuinely sub-perceptual. Under pointwise image-only inspection, G2–G6 sit at chance for 4B and 8B models, and a 30B model breaks only the coarsest class (G1 overflow). This floor is invariant to the things one would tune first: swapping the vision encoder across *five* families (SigLIP2, an LLM-based encoder, InternViT, a

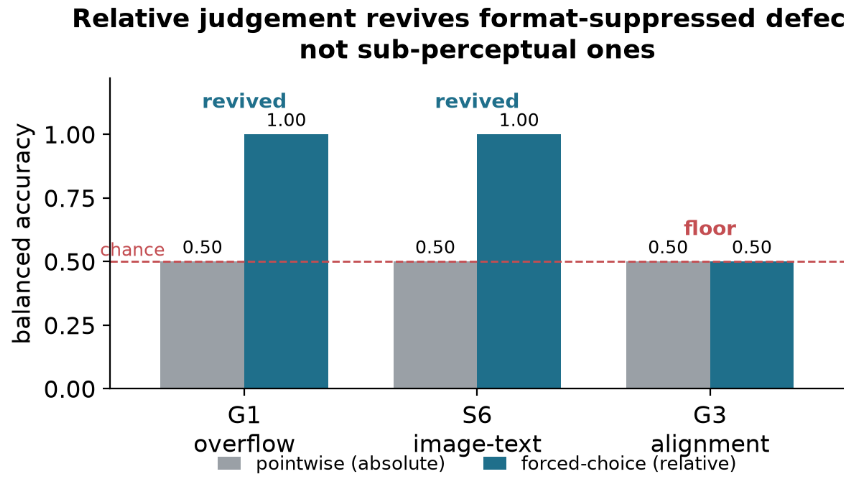

Figure 2: Relative judgement separates the two blindnesses. For G1 overflow and S6 image/text contradiction, pointwise (absolute) inspection is at chance but a forced choice recovers perfect detection (*elicitation-recoverable*). For G3 alignment, the forced choice stays at chance (*genuinely sub-perceptual*). Same model, same images; only the elicitation changes. What *drives* the recovery (naming vs. a paired reference) is decomposed in Fig. 4.

native-resolution NaViT, and MoonViT) leaves every model at the random line, differing only in abstain-vs-over-flag bias, and raising input resolution from 1536 to 2048 changes nothing. The lever is model scale and the elicitation, not the encoder or the pixel budget—so for fine geometry the right detector is the symbolic linter, which reads the coordinates directly at balanced accuracy 0.75–1.0 and zero false positives.

But some "perceptual" failures are elicitation artifacts, not blind spots. The same pointwise image-only setting hides a second, opposite phenomenon. For G1 text overflow and S6 image/text contradiction, pointwise balanced accuracy is 0.50—indistinguishable from the genuinely sub-perceptual classes—yet a two-alternative forced choice between the defective and clean slide jumps to **1.00** (Fig. 2): the defect is perceivable, and the absolute pointwise call simply fails to elicit it. Crucially, the same forced-choice intervention does *not* rescue G3 alignment (it stays at 0.50): relative judgement is a real effect with a sharp boundary, and that boundary is exactly the perception/elicitation split. (*How* the elicitation helps differs by class—a named target on a single slide vs. a paired clean reference—and we decompose it in Sec. 5.1 (Fig. 4), finding pure format suppression for the constructed render class G7 and a milder availability-of-reference effect for G1/S6.) This recoverable-versus-sub-perceptual boundary—two independent instances, one geometric, one semantic—is the connective thread of the paper.

One honest counterexample: terminology. Not every consistency defect is format-suppressed. S3 terminology inconsistency is *not* rescued by forced choice (0.625, heavy position bias): reading the same term across deck pages and spotting a subtle variant is below threshold even side by side. Tellingly, the structured-oracle channel—which feeds the model the full deck text with both variants present—also fails (0.56), so the bottleneck is OCR-from-pixels plus the yes/no framing, not reasoning. S3 therefore leaves the VLM entirely for a symbolic terminology linter, which reaches balanced accuracy 1.000 against the VLM's 0.69. A negative result that *re-routes* a class is exactly the kind the diagnosis is for.

A methodological aside that becomes load-bearing. Enterprise "snap-to-master" templates re-fit element boxes to a grid before rendering. Measuring this, we find a template silently *absorbs* 45% of injected geometry defects: overlap, alignment, and margin violations become pixel-identical to their clean twins after snapping (Sec. 5). This collapses the inspectable spectrum toward two poles—pure-perception defects that templates cannot constrain (overflow, driven by content-determined text length) and pure-semantic defects that templates do not touch—and it is a trap for any dataset that derives labels from the IR but evaluates on the rendered image. We use it below to explain a reward-model failure and to justify rendering defectives faithfully.

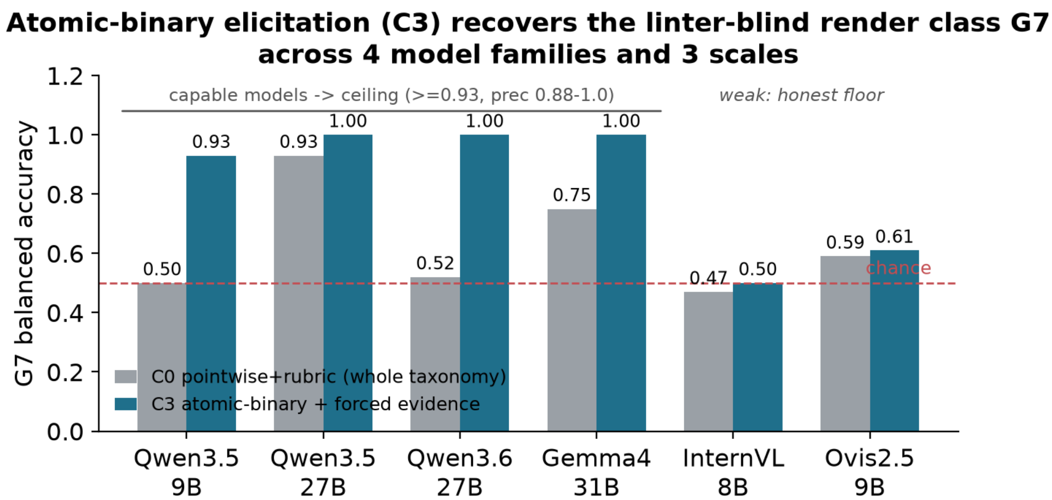

Figure 3: Atomic-binary elicitation (C3) recovers the linter-blind render class G7 across families. Capable models reach the ceiling (≥0.93, precision 0.88–1.0); the weaker InternVL/Ovis stay near their perceptual floor—an honest, capability-dependent boundary, not a universal fix.

5 A routed symbolic–neural critic

The diagnosis says a single model is the wrong unit: route each defect to the engine that owns its bottleneck. We build that critic, validate the elicitation recovery it relies on across many models, and stress it with the constructed render class G7.

5.1 Elicitation: format suppression is not capability

Four ways to ask. We compare four elicitations holding the model and image fixed: **C0**, the conventional pointwise+rubric pass over the whole taxonomy in one call; **C1**, free-form describe-then-classify; **C2**, a geometry-normalized synthetic-twin pairwise (re-render snapped-to-grid, both orders); and **C3**, an atomic per-type yes/no with forced localization. The scientific contrast is **C3 vs. C0**: same model, taxonomy, and image, differing only in "ask everything at once" versus "one atomic binary with forced evidence." If C3 $\gg$ C0, the pointwise abstention was a format/cognitive-load artifact, not a capability gap.

The recovery replicates across families and scales. On the constructed render class G7, the atomic-binary C3 recovers detection that C0 suppresses, and it does so across four model families and three scales (Fig. 3): Qwen3.5-9B $0.50 \to 0.93$, Qwen3.5-27B $0.93 \to 1.00$, Qwen3.6-27B $0.52 \to 1.00$, and Google Gemma4-31B $0.75 \to 1.00$, all at precision 0.88–1.00. As a *cross-model* effect—the claim we actually make—the recovery is overwhelming: a stratified (Cochran) McNemar over the four models' matched pairs counts **149** G7 items that C3 fixes against **zero** reversals (χ^2_1=149, $p < 10^{-30}$).[1] That the *same* effect appears in models from different vendors with different pretraining is the core model-agnostic evidence: C0's fixed taxonomy cannot *name* an off-taxonomy render class, so it abstains; the atomic per-type binary restores it.

It is real perception, not a lucky "yes." Paired-clean precision near 1.00 already rules out an always-flag model, but to confirm a correct "yes" points to the right place we check C3's forced evidence against the synthetic G7 ground truth. On the four capable models' G7 true positives, the named region and the named overflowing *element* are correct ≥98%; models read back the specific spilling content ("the list items starting from 'Cost attribution' "), which a bluffing model cannot do. The recovery is perception, not a calibration trick on the label.

[1]Per model the gain is individually significant for the three low-C0 models ($p \leq 2\times10^{-9}$); on the already-strong Qwen3.5-27B (C0 0.93) only 9 items can move and all 9 go to C3 (0 reversals, exact p=0.004, individually n.s. after Holm over the full 59-test family)—a ceiling effect, not a fragile one. Holm (FWER) over all 59 reported tests keeps the pooled G7 recovery, the three low-C0 G7 cells, both examiner contrasts (p_{Holm}=1.9×10^{-5}, 4.7×10^{-4}) and the Skywork G7 detection; full corrected table + the stratified test in the released multiplicity report.

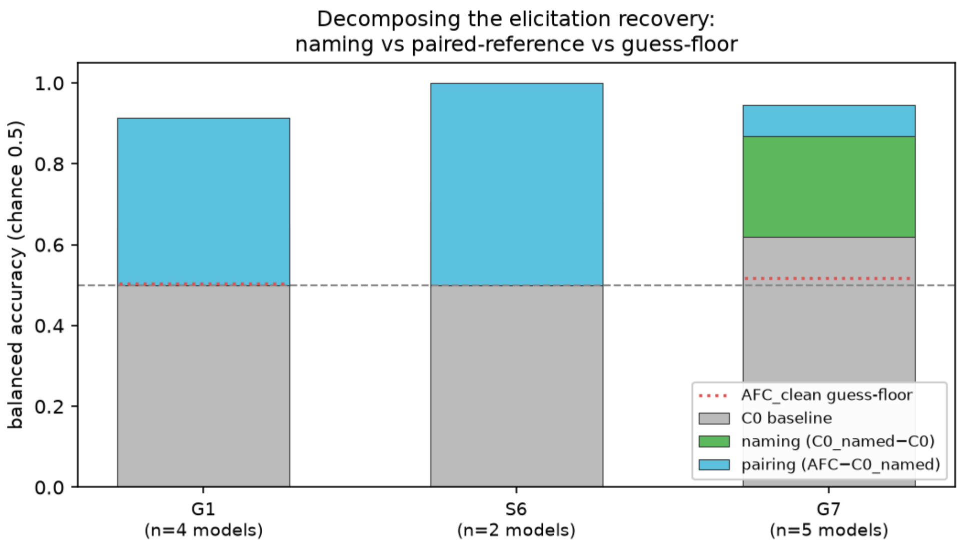

Figure 4: What drives the recovery: naming vs. a paired reference. Holding model and image fixed on the freeform set, the chance→ceiling jump factors into a *naming* component (C0_named−C0: one slide, the defect named, *no* reference) and a *pairing* component (AFC−C0_named: a clean twin supplied). For the constructed render class G7 naming alone carries it (green); for declared geometry G1 and semantic S6 the paired reference does (blue). The AFC_clean guess-floor (dotted) is near chance, so neither is a forced-pick artifact. Mean over the models that recover (AFC $\geq$ 0.7).

No free lunch: elicitation is conditional. The optimal elicitation is jointly determined by model capability and defect class. C2 synthetic-twin pairwise—not C3—is what rescues *declared* geometry G1 on capable models (Qwen3.5-27B 0.69 → 0.93, Qwen3.6-27B 0.53 → 0.78, precision 1.00), because snapping absorbs the declared overflow and the pairwise contrast exposes it. And C1 free-form, which rescues weak abstaining models, *collapses* on strong models: the capable, verbose models free-associate nits and flag a defect on nearly every clean slide (precision ≈0.50). There is no universal elicitation—which is precisely why the critic routes per defect and per model rather than adopting one prompt.

Naming vs. a paired reference: a defect-dependent answer. A skeptic asks whether the chance→1.00 recovery is genuine *elicitation* (the model perceives the defect and the format suppressed it) or merely an *easier task* (the forced choice hands over a clean reference and a named target the pointwise call withholds). We factor it, holding model and image fixed, into a *naming* term (C0_named: a single slide, the candidate defect named, no reference, no forced-evidence gate) and a *pairing* term (the true 2-AFC against the clean twin), with a clean-vs-clean *guess-floor* control (Fig. 4). The answer is *defect-dependent* and consistent across all four capable models. For G7—the off-taxonomy render class C0 cannot *name*—**naming alone** recovers it (Qwen3.5-9B 0.50 → 0.93, Qwen3.6-27B 0.51 → 1.00, Qwen3.5-27B 0.79 → 1.00, Gemma4-31B 0.69 → 0.88; paired-clean McNemar Holm-corrected $p \leq 0.002$), and the paired reference adds nothing on top: this is *format suppression*, the model saw G7 all along. For G1/S6 the opposite holds: naming alone does *not* help (C0_named ≈0.50) and the recovery needs the paired clean reference (2-AFC →0.97–1.00)—an *availability-of-reference* effect, properly scoped as such. Crucially, the guess-floor control rules out a forced-pick artifact for both: on two *clean* slides the capable models correctly answer "tie" 92–100% of the time and fabricate a consistent winner on ≤2% of pairs, so the 2-AFC ceiling is real signal, not always-pick-one. The paper's mechanism claim—*format suppression, not the eyes*—is thus established for the constructed G7 (and the C3 atomic recovery it mirrors); the declared G1/S6 forced-choice recovery is the milder, honestly-labelled availability-of-reference effect.

It transfers to real slides. On the human-annotated SlideAudit set, even with no structure available (so no linter), C3 beats C0 pointwise for every class—brand color 0.52 → 0.81, alignment 0.56 → 0.72—confirming the format effect is not a synthetic artifact. Absolute geometry on bare real pixels stays modest (0.55–0.74 outside blatant overlap): real fine geometry still needs the linter, which needs the element structure bare images lack. The hybrid's full power requires native structure; on image-only real data it runs in a degraded VLM-only mode—a ceiling set by missing structure, not missing method.

Table 2: Per-class named-attribution balanced accuracy with 95% Wilson intervals. E3 adds C3-everywhere and linter+VLM-C3: the minimal hybrid matches routed coverage and slightly exceeds its mean. n = paired positives with matched clean controls; precision for every cell is in the released reports.

Class	routed to	linter	VLM-C0	VLM-C3	linter+C3	routed	n
G1 overflow	linter	1.00 [.91,1]	0.50 [.46,.54]	0.50 [.46,.54]	**1.00 [.91,1]**	**1.00 [.91,1]**	40
G2 overlap	linter	0.80 [.68,.87]	0.71 [.60,.79]	**0.86 [.74,.92]**	0.80 [.68,.87]	0.80 [.68,.87]	40
G3 alignment	linter	1.00 [.90,1]	0.47 [.41,.54]	0.50 [.45,.55]	**1.00 [.90,1]**	**1.00 [.90,1]**	36
G5 color	linter	1.00 [.91,1]	0.50 [.46,.54]	0.50 [.46,.54]	**1.00 [.91,1]**	**1.00 [.91,1]**	40
G6 margin	linter	1.00 [.90,1]	0.50 [.45,.55]	0.50 [.45,.55]	**1.00 [.90,1]**	**1.00 [.90,1]**	36
G7 render-overflow	VLM (C3)	0.00 [0,.54]	0.50 [.46,.54]	**1.00 [.91,1]**	**1.00 [.91,1]**	**1.00 [.91,1]**	40
S1 title/body	VLM (C0)	0.50 [.41,.59]	**0.94 [.75,.98]**	0.83 [.63,.92]	0.83 [.63,.92]	**0.94 [.75,.98]**	18
S4 density	LLM	0.50 [.45,.55]	0.51 [.44,.59]	**0.93 [.81,.97]**	**0.93 [.81,.97]**	0.72 [.60,.80]	36
S6 image/text	VLM	0.50 [.41,.59]	0.50 [.41,.59]	0.50 [.41,.59]	0.50 [.41,.59]	0.50 [.41,.59]	18
mean / classes covered		0.70 / 5/9	0.57 / 2/9	0.681 / 4/9	**0.896 / 8/9**	0.885 / 8/9	

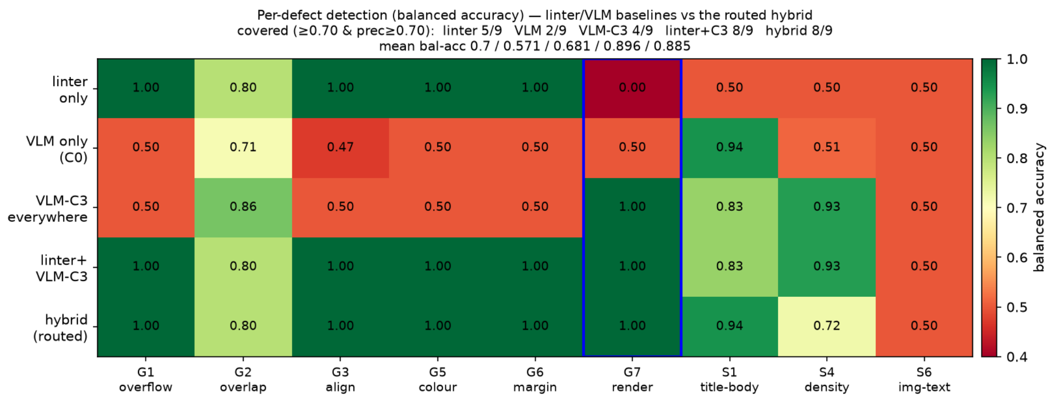

Figure 5: Coverage by engine. Per-defect balanced accuracy for the linter, VLM-C0, VLM-C3, linter+VLM-C3, and the routed hybrid. Minimal linter⊕C3 matches routed coverage; G7 (boxed) is the linter-blind cell rescued by C3.

5.2 The minimal hybrid critic covers the most

Static routing, complementary engines. We evaluate five critics by *named attribution*: the correct defect must be emitted on the defective slide and not on its clean twin. The E3 control is decisive: the minimal hybrid—linter for declared geometry plus one C3-prompted VLM elsewhere—covers **8/9** classes at **0.896**, matching the pre-registered routed hybrid (**8/9**, **0.885**) and beating either single engine (linter: **5/9** at 0.70; VLM-C3: **4/9** at 0.681; VLM-C0: **2/9** at 0.57; Fig. 5, Table 2). The nine classes are the image-level paired-clean classes; G4/S3 are linter-owned and S2/S5 are deck-scope (Sec. 6). Thus the falsification branch strengthens the message: per-class routed prompt/LLM tuning is not load-bearing here; *linter ⊕ C3-VLM* is the cleaner result.

A data-driven routing correction, reported as such. We initially routed S1 title/body mismatch to the text LLM, but the text-only probe over-flags (0.25, precision 0.09) while the image-bearing VLM names it reliably (0.94, precision 0.90): title/body mismatch needs the *rendered layout*, not just the text. We re-route S1 to the VLM and report it as a refinement, not a tuned result. We also do not hide the one class the hybrid fails to cover: S6 image/text contradiction stays at 0.50 for every engine.

5.3 G7: a falsifiable class that the linter and narrow rewards miss

The construction. G7 is defined to make the linter's blind spot falsifiable: a declared box that is legal (in-margin, non-overlapping) but whose rendered content overflows its container. The IR carries only legal short text; the overflow

Table 3: Multi-reward audit: paired preference accuracy $P(r_{\text{clean}} > r_{\text{def}})$ per class (chance = 0.5; G7 with 95% CI). Three published scorers spanning document, general-multimodal, and aesthetic categories; each is sensitive to several in-taxonomy classes. G7 (boxed in Fig. 6) splits them: the narrow document and aesthetic rewards sit at chance (CI spans 0.5), only the general-multimodal reward detects it.

Reward scorer	category	G1	G2	G5	G7	S4	S6
DocReward-3B	document	0.93	1.00	0.37	**0.48** [.38, .58]	1.00	1.00
Skywork-VL-7B	general-mm	0.94	0.72	0.46	**0.79** [.69, .86]	0.92	1.00
LAION-Aesth.	aesthetic	0.37	0.91	0.57	**0.57** [.46, .66]	0.75	0.61
n (paired)		54	54	54	90	36	18

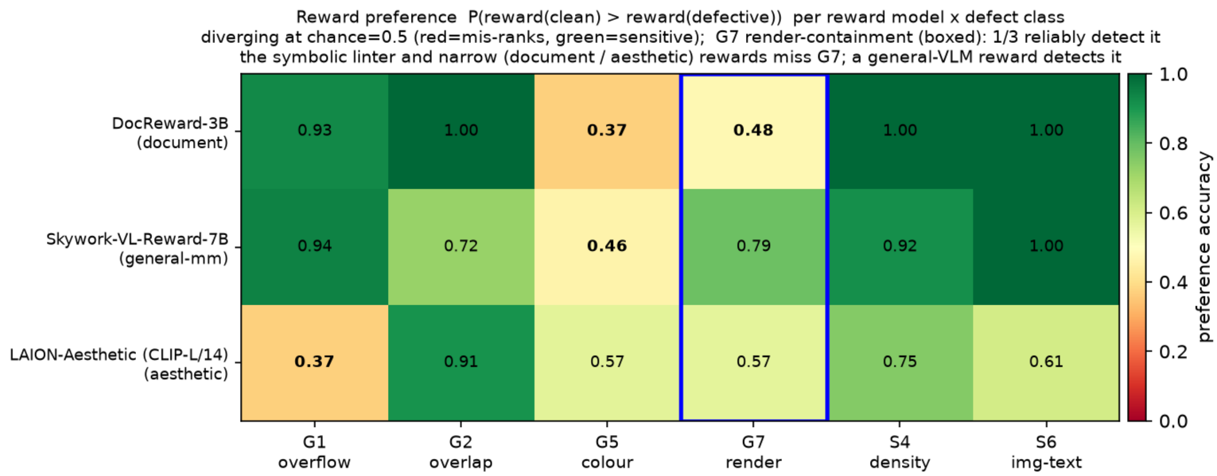

Figure 6: Narrow rewards miss G7; a general-multimodal reward catches it. Preference accuracy per reward model × defect class (green = sensitive, red = mis-ranks; chance = 0.5). Every scorer is sensitive to several in-taxonomy classes, but for the boxed G7 render class the document (DocReward 0.48) and aesthetic (LAION 0.57) rewards are at chance while the general-multimodal Skywork-VL (0.79) detects it—so G7 is a blind spot of *narrow* critics, caught by a capable VLM (prompted or reward-headed).

lives in the renderer, not the IR, so the linter and any structure-only model are blind by construction. A self-check confirms the geometry linter returns zero findings on 90/90 G7 defectives.

Narrow neural rewards are also blind; a general-multimodal reward is not. The natural rebuttal is "just train a neural reward model." We test it directly by auditing *three* published reward scorers spanning three categories and backbones on the same paired slides—does each score the clean slide above its defective twin? **DocReward** [11] is a Qwen2.5-VL-3B *document* reward with a Bradley–Terry head trained on 117K paired documents; **Skywork-VL-Reward-7B** [12] is a Qwen2.5-VL-7B *general*-multimodal value-head reward; and **LAION-Aesthetic** [13] is an *aesthetic* linear head on CLIP-L/14. Each cleanly penalizes the in-taxonomy defects it can see (preference accuracy 0.72–1.00 on overflow, overlap, density, image/text), so each is a working reward, not a domain mismatch (Table 3, Fig. 6). On G7 the picture *splits*: the symbolic linter (0.00), the document reward (DocReward 0.48, 95% CI $[0.38, 0.58]$) and the aesthetic scorer (LAION 0.57, $[0.46, 0.66]$) all sit at chance—their CIs include 0.5, so they carry no reliable signal about the render-containment overflow—while the general-multimodal reward (Skywork-VL 0.79, $[0.69, 0.86]$) *does* catch it, its Qwen2.5-VL-7B backbone perceiving the overflow even without being told to look. So the G7 blind spot is *not* model-agnostic across all neural rewards; it is a property of *narrow* critics—a rule-based linter and specialised document/aesthetic reward heads—whereas a capable general VLM catches it. This is the neural counterpart of the hybrid thesis: G7 needs a capable VLM engine, and that engine works whether it is *prompted* (the C3 recovery above, 0.93–1.00) or carries a *reward head* (Skywork 0.79). The honest revision of a single-model audit is therefore "narrow rewards are blind to G7; a general VLM reward, like a re-elicited VLM, is not"—a reviewer who tests Skywork will find 0.79, and we report it. (Naming containment in Skywork's prompt lifts G7 to 0.87, reinforcement rather than rescue; DocReward is meanwhile also blind to brand-color G5 at 0.37, a rule class the linter owns at 1.00.)

Table 4: Page-semantic inspection, balanced accuracy [95% Wilson] on paired clean (best modality). The finetuned 8B examiner beats both zero-shot baselines, including the 30B, *in-distribution*. Deck-scope S2/S5 are an honest negative (text). n = paired positives (2-AFC rows = forced-choice pairs); the S6 2-AFC rests on only 12 pairs (wide interval).

class	**ft-8B**	zs-8B	zs-30B	n
S1 title/body	**1.000** [.95,1]	0.778 [.69,.85]	0.933 [.87,.95]	36
S4 density	**1.000** [.97,1]	0.529 [.50,.58]	0.774 [.69,.84]	60
G1 overflow (2-AFC)	**1.000** [.97,1]	0.975 [.93,.99]	0.700 [.61,.77]	120
S6 image/text (2-AFC)	**1.000** [.76,1]	1.000 [.76,1]	0.500 [.25,.75]	12
S2 narrative (deck)	0.500† [.45,.53]	0.646 [.53,.72]	0.549 [.42,.66]	36
S5 missing-logic (deck)	0.500 [.47,.53]	0.500 [.47,.53]	0.567 [.45,.68]	60

† degenerate (always-flag): the SFT mix had no clean-*deck* negatives. See text.

Perturbation fidelity: 45% of injected geometry never renders. The reward audit ties to the template result of Sec. 4. Rendering every injected defective two ways—faithfully and snapped-to-master—we find 45% of injected geometry defects (100% of overlap, alignment, margin; 20% of overflow) become pixel-identical to their clean twin after snapping. Feeding the *snapped* pairs to *all three* reward scorers gives preference accuracy 0.00 at a reward gap of exactly 0.000: the two images are pixel-identical, so no critic—document, general-multimodal, or aesthetic—can react—a dataset-integrity hazard (identical inputs force identical scores), not a reward-model deficiency. This part *is* model-agnostic by construction, and any reward model trained or evaluated on snap-rendered images with IR-derived labels inherits these zero-signal pairs. Perturbation fidelity must be checked at the pixel level, not assumed from the IR—a prerequisite the field largely skips, and the reason our elicitation experiments render defectives faithfully.

6 The semantic engine: a specialist examiner

The hybrid's semantic engine can be a zero-shot model, but a small specialist does better on its own ground. We finetune Qwen3-VL-8B with QLoRA on programmatically injected defects (2,142 routed SFT records: semantic pointwise, geometry restate-from-structure with image-only abstention, and G1/S6 pairwise), and evaluate on held-out splits.

An 8B examiner surpasses a zero-shot 30B on *in-distribution* semantics. On in-domain held-out page-semantic inspection, the finetuned 8B reaches balanced accuracy 1.000 on both S1 title/body and S4 density, beating zero-shot 8B (0.78 / 0.53) and zero-shot 30B (0.93 / 0.77) (Table 4). The largest gain is S4 density recall, $0.0 \to 0.97$, where the finetuned 8B (1.00) beats the zero-shot 30B (0.65) at two-proportion z, $p = 4.5 \times 10^{-7}$: an 8B examiner surpasses a 4× larger zero-shot model on the examiner's core job. The advantage is not a prompt-format artifact—re-running the zero-shot baselines at the finetuned model's own trained prompt format collapses them to 0.50, so the format asymmetry favors the baselines, and the finetuned model still wins at each model's best format. On held-out *severities* the semantic edge holds (0.89 vs. 0.50–0.59).

It abstains instead of hallucinating geometry. Under image-only inspection, the finetuned model stays at 0.50 on G2–G6 with *zero* false positives: trained with image-only abstention targets, it declines to hallucinate geometry from pixels and leaves those classes to the linter—the designed behavior, and the reason the hybrid's geometry column comes from the linter rather than the VLM.

Honest negatives. Two results cut against an over-broad reading. First, *deck-scope* semantics fail pointwise: the finetuned model is degenerate on S2 (recall 1.0, specificity 0.0—always flags) because the SFT mix contained no clean-*deck* negatives, and it abstains on the held-out S5; deck-scope classes were never given the pairwise head the diagnosis prescribes for this failure mode, and fixing this needs a retrain we did not run. Second, there is a *sim2real gap*: on image-only SlideAudit, the finetuned model's synthetic S4 strength (0.97) does not transfer (0.50); only the zero-shot 30B shows any real image-only density signal (recall 0.53 vs. the finetuned 0.075, $p = 1.1 \times 10^{-5}$). Scale, not finetuning, carries bare-pixel transfer—and the finetuned model learned a defect *distribution*, not a universal notion of slide quality. We make this concrete next.

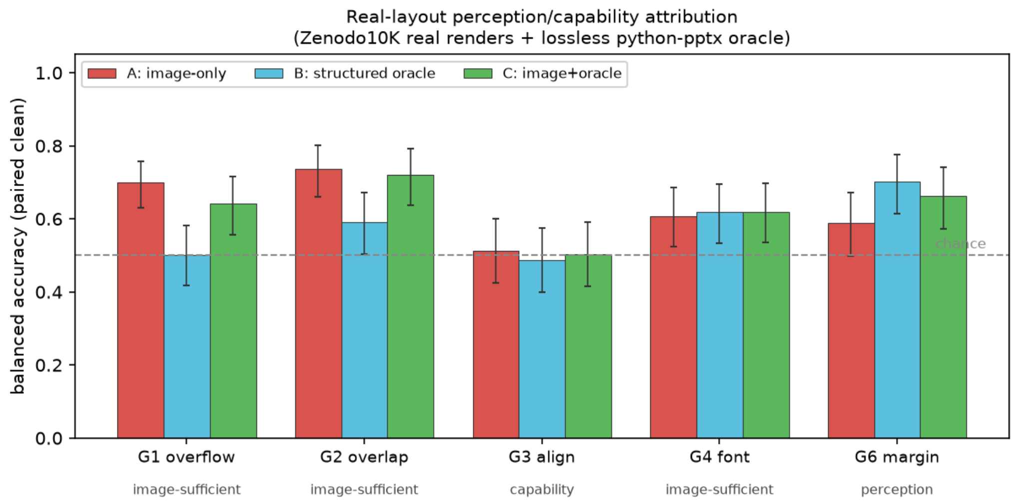

Figure 7: The perception/capability split reproduces on real layouts (Zenodo10K real renders + lossless `python-pptx` oracle; mean over 3 VLMs / 3 families, balanced accuracy on paired clean; error bars = 95% Wilson on counts pooled across models, n=76–90 paired per class). Coarse defects (G1 overflow, G2 overlap) are image-sufficient; G6 margin is a *perception* bottleneck the structured oracle rescues (B>A); G3 alignment is a *capability* floor at chance in every channel (A=B=C≈0.5), so it must route to the linter. Intervals on G3/G4/G6 include chance, consistent with their small per-class n (see released reports).

7 External validity

A real pre-sales agent deck. We run the same five-level examiner gradient on a real 16-slide proposal produced by a deployed PowerPoint agent. Its verifiable defect is 20 unfilled template placeholders ("add a title here") across 7 of 16 slides—a semantic-completeness defect the geometry linter is blind to by construction (the boxes are geometrically fine). The result reinforces the thesis from an unexpected angle: the zero-shot 30B is the best detector (recall 1.0, names the placeholder on 4/7 slides, scores them 0.03 vs. 0.64 for clean), while the *finetuned* 8B—top of the intrinsic-quality ranking—is among the *worst* on this out-of-distribution defect (names it 0/7), and the linter's apparent recall is spurious geometry noise with zero placeholder mentions. The ranking on a novel defect is by general-VLM ability, not by the specialist's in-distribution quality: the examiner learned a distribution, not universal quality. Feeding the flagged placeholders back to the agent's own generator for one revision pass drives the verifiable defect from 20 to 0.

The structured attribution holds on real layouts. The body runs the A/B/C attribution on synthetic slides; the cleanest discrimination needs it on *real* geometry with a *real* oracle, which seemed to require human or model annotation. We obtain it without either: Zenodo10K [14] ships real CC-licensed `.pptx` with intact element XML, so `python-pptx` extracts a geometry/text/style oracle *lossless with respect to the declared IR* straight from the source file (the IR→render gap that G7 exploits is exactly what this oracle does *not* capture, by design)—no human, no self-annotation bias. From 26 real decks we inject one single-shape defect per slide *in PPTX XML space* (overflow, overlap, alignment offset, font, margin) and render both the clean and the defective one-slide deck with the same real renderer (LibreOffice), so the paired pixel diff isolates the injected defect on real pixels (209 paired slides, 5 classes, 3 VLMs / 3 families). Two results follow. First, the §4 "snap absorbs 45%" hazard is *template-specific*: real free-form decks absorb only **7%** (render-fidelity 0.93), so the perturbation is faithfully present—a positive control on that claim. Second, *all three* attribution outcomes appear on real geometry: coarse defects (overflow, overlap) are **image-sufficient** (balanced accuracy A 0.70–0.74, structure adds nothing); a margin bleed is a **perception bottleneck the oracle rescues** for the weaker VLMs (A 0.59 → B 0.70)—the exact "missing structure, not missing capability" cell; and fine alignment is at chance in *every* channel for *every* model (A = B = C ≈ 0.50), a genuine capability floor the VLM-consumed oracle does not cross, confirming on real geometry that fine geometry must route to the symbolic linter rather than a VLM (Fig. 7).

Does a better examiner help downstream? A small, honest effect. We also asked whether examiner quality transfers to generation: vary the examiner across five intrinsic-quality levels, feed its critique into a self-refine loop, and measure an independent model-free quality. The direction is the hypothesized one—examiner quality correlates +0.66 with refinement gain, ranking the linter last and the strong semantic examiners first—but the magnitude is sub-1% because a mainstream generator floors most briefs at the first draft, leaving little headroom; a second vehicle (skill-space optimization) even reverses sign as an artifact of its efficiency metric. We report this as a bounded, honest negative on *magnitude*, not as a downstream-utility claim: in this floored regime the examiner-quality-to-utility signal is real but small, and the verifiable evidence for the examiner is its intrinsic and case-study performance, not the downstream loop.

8 Limitations

We *do* run the structured (image+oracle) evaluation on real slides that separates "missing structure" from "missing capability" on real geometry (§7): a lossless `python-pptx` oracle over real CC-licensed decks sidesteps both the annotation labour and the self-annotation bias, and the split reproduces—fine alignment is a capability floor in every channel while a margin bleed is structure-rescuable. Its scope is honest: the defects are *injected* (real geometry, real renderer, real oracle, but not naturally-occurring), so the third-party SlideAudit set (image-only, naturally-defective) remains our complementary real probe. We audit three published reward scorers (document, general-multimodal, aesthetic); two further rewards are downloadable but need runtimes we did not stand up (an InternLM-family reward whose PLoRA image path failed to load, and a design reward needing a separate training runtime). Our attribution protocol shares its core with concurrent perception/reasoning work, differentiated by domain, the lossless oracle, and the loop to a critic. And the downstream effect is small under a strong generator. None of these is hidden; each is stated where it bites.

Two further caveats temper the central claim and we state them plainly. First, the forced-choice and atomic-binary elicitations supply a contrastive reference and a named target that the pointwise format withholds, so one must ask whether the $0.50 \rightarrow 1.00$ recovery is *elicitation* or *reduced task difficulty*. We ran the clean ablation rather than leaving it to future work (the single-slide, defect-type-named *absolute* condition C0_named, plus a clean-vs-clean guess-floor control; Sec. 5.1, Fig. 4), and the answer is defect-dependent: for the constructed render class G7 naming alone recovers detection (format suppression), while for declared G1/S6 the recovery genuinely requires the paired reference (availability-of-reference). We therefore make the strong mechanism claim only for G7 and its C3 mirror, and label the G1/S6 forced-choice effect as reference-availability; the guess-floor control (capable models answer "tie" on 92–100% of clean pairs) rules out a forced-pick artifact for both. Second, the 8/9 coverage (0.896 for linter+VLM-C3; 0.885 for the pre-registered routed hybrid) is a *native-structure* number; on image-only real decks (no IR for the linter) the hybrid degrades to its VLM-only mode, so that figure is an upper bound for IR-owning generation agents, not an open-world guarantee.

9 Conclusion

A slide-defect critic should not be one model. Telling apart the two blindnesses—genuinely sub-perceptual geometry versus format-suppressed perception—turns a vague "VLMs are bad at layout" into an actionable routing rule: rules to the linter and perceivable-but-suppressed classes to a VLM under an elicitation that the diagnosis proves recovers them. The E3 control makes the rule simpler: linter plus one C3-prompted VLM covers 8/9 classes, matching the pre-registered routed hybrid and beating either single engine. A falsifiable render class G7 shows that the linter's blind spot is shared by *narrow* neural rewards (document, aesthetic) but not by a capable general-multimodal reward—so G7 needs a VLM engine, prompted or reward-headed, not a narrow critic. The recovery replicates across four of the six VLMs tested and four families and transfers to real slides, which makes the central claim a property of the *elicitation* × *defect* interaction rather than of any one model. The unit of contribution is the diagnosis-to-routing pipeline and the falsifiable class, not a leaderboard number—and the negatives we kept in are part of the evidence.

Reproducibility

All analysis scripts, the defect injector, the two linters, the elicitation harness, the hybrid router, the multi-reward audit adapters, the real-layout (Zenodo10K) IR/inject/render pipeline, and the per-Part reports are released. Experiments ran on 4×RTX 3080 (20 GB) with vLLM-served open VLMs (Qwen3.5/3.6, Gemma4, InternVL3.5, Ovis2.5), a QLoRA-finetuned Qwen3-VL-8B, and three published reward scorers (DocReward-3B, Skywork-VL-Reward-7B, a LAION aesthetic predictor) behind a uniform adapter; serving recipes (tensor-parallel, AWQ, disabled thinking, fp8-KV caveats on Ampere) are in the repository. Weights, renders, and raw rollouts are not redistributed. Every balanced-accuracy cell in the tables and figures carries its 95% Wilson interval and per-cell n; the full family of paired tests is reported with Holm (FWER) and Benjamini–Hochberg (FDR) multiplicity correction over all 59 reported significance tests (released multiplicity report), and headline claims are stated as surviving or honestly downgraded under that correction.

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